Phase 1 Report

Applied Machine Learning Baseline

Automatic Clothing Category Recognition on Fashion-MNIST

Course Project 11 | Team T112 | Section D | October 2026

#	Member	GitHub	Contribution
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1. Problem Statement

Commercial inventory systems must reliably tag thousands of clothing images per day. Fashion-MNIST provides a standardized 10-class benchmark (70,000 grayscale 28x28 images) representative of this task. Phase 1 evaluates four classical ML algorithms on two visual representations to establish a rigorous baseline and identify the exact failure modes that motivate a Phase 2 Convolutional Neural Network.

Primary research question: Can fixed, handcrafted visual descriptors adequately classify clothing for a commercial inventory catalogue, or does fine-grained category confusion necessitate hierarchical visual representations learned by a CNN?

2. Experimental Setup

2.1 Dataset and Data Splits

Split	Size	Purpose
Training (dev)	48,000	Model selection & hyperparameter tuning
Validation	12,000	Held-out evaluation during tuning
Test (official)	10,000	Final one-time evaluation (held-out)

All splits use stratified sampling with random_state=42. Pixel values are normalized from uint8 [0, 255] to float32 [0.0, 1.0].

2.2 Class Labels

ID	Class	ID	Class
0	T-shirt/top	5	Sandal
1	Trouser	6	Shirt (most challenging)
2	Pullover	7	Sneaker
3	Dress	8	Bag
4	Coat	9	Ankle boot

2.3 Feature Representations

Feature	Dimensions	Description
Pixels	784-d	Flattened 28x28 normalized pixel values in [0.0, 1.0]
HOG	324-d	9 orientations, 7x7 px/cell, 2x2 cells/block, L2-Hys normalization

2.4 Models and Best Hyperparameters

Model	Best Config (Pixels)	Best Config (HOG)
Logistic Regression	C=0.1, solver=lbfgs	C=0.1, solver=lbfgs

Model	Best Config (Pixels)	Best Config (HOG)
k-NN	k=7, weights=distance	k=5, weights=distance
Random Forest	200 trees, max_depth=20	200 trees, max_depth=None
Linear SVM	C=0.1, max_iter=2000	C=1.0, max_iter=2000

2.5 Evaluation Metrics

- **Accuracy** -- fraction of correct predictions on the 10,000 test images.
- **Macro F1** -- unweighted mean F1 across all 10 classes.
- **Per-class Recall** -- sensitivity per garment category.
- **Parameter Count** -- weights for linear models, MiB for k-NN, nodes for RF.
- **Inference Latency** -- median of 5 warmed-up calls on 1,000-image batches (ms/image).

3. Results

3.1 Master Comparison Table

#	Model	Features	Test Acc	Macro F1	Parameters	Latency (ms)
1	**Random Forest**	**Pixels**	**87.43%**	**87.31%**	**1,693,570 nodes**	**0.0264**
2	k-NN	Pixels	85.41%	85.40%	179.4 MiB	0.2741
3	Logistic Regression	HOG	84.55%	84.49%	3,250	0.0004
4	Random Forest	HOG	84.35%	84.30%	2,458,016 nodes	0.0291
5	Logistic Regression	Pixels	84.32%	84.24%	7,850	0.0007
6	Linear SVM	HOG	84.23%	84.06%	3,250	0.0006
7	k-NN	HOG	83.84%	83.85%	74.2 MiB	0.0980
8	Linear SVM	Pixels	83.80%	83.57%	7,850	0.0013

Table 1. All 8 model configurations ranked by test accuracy. Best configuration highlighted (Random Forest + Pixels, 87.43%).

3.2 Key Findings

Finding 1: Random Forest dominates overall. RF on pixels achieves 87.43% accuracy, outperforming k-NN by +2.02 pp, Logistic Regression by +3.11 pp, and Linear SVM by +3.63 pp.

Finding 2: HOG helps linear models, hurts non-linear ones. For LogReg and SVM, HOG improves accuracy by +0.23-0.43 pp while reducing parameter count by 58.6% (7,850 -> 3,250). For RF and k-NN, HOG reduces accuracy by 3.08 pp and 1.57 pp.

Finding 3: k-NN memory and latency trade-off. Switching from pixels (179.4 MiB) to HOG (74.2 MiB) reduces k-NN memory by 58.7% and inference latency by 2.4x at only 1.57 pp accuracy cost.

Finding 4: Linear classifiers are fastest. Logistic Regression on HOG runs at 0.0004 ms/image, 685x faster than k-NN on pixels.

Finding 5: Shirt (class 6) is universally hardest. Mean Shirt recall across all 8 configurations is 57.7%. No model exceeds 62.6%.

4. Per-Class Recall Analysis

4.1 Best Model -- Random Forest (Pixels)

Class	Label	Recall	Assessment
0	T-shirt/top	85.9%	Good
1	Trouser	96.1%	Excellent
2	Pullover	78.9%	Fair
3	Dress	90.5%	Good
4	Coat	81.1%	Good
5	Sandal	95.6%	Excellent
6	Shirt	59.3%	Poor
7	Sneaker	95.2%	Excellent
8	Bag	97.3%	Excellent
9	Ankle boot	94.4%	Excellent

Table 2. Per-class recall for the best model (Random Forest, Pixels, 87.43% accuracy).

4.2 Cross-Model Shirt Recall

Model	Pixels	HOG
Logistic Regression	57.1%	57.8%
k-NN	60.2%	62.6%
Random Forest	59.3%	59.4%
Linear SVM	50.7%	54.6%
Mean	56.8%	58.6%

Table 3. Shirt (class 6) recall across all 8 configurations. No model exceeds 62.6%.

5. Error Analysis: The Shirt Dilemma

5.1 Confusion Pattern

Shirt (class 6) is the most visually ambiguous upper-body label. Its 28x28 silhouette overlaps with T-shirts (similar collar and sleeve length), Pullovers (similar torso coverage), and Coats (similar outer garment outline). The confusion is directional:

Error Pattern	Frequency
Shirt predicted as T-shirt/top	~15-16% of Shirt test images
Shirt predicted as Pullover	~12-14% of Shirt test images
Pullover predicted as Coat	~8-10% of Pullover test images

5.2 Why Fixed Representations Fail

Flattened Pixels (784-d)

Linear classifiers assign one weight per pixel coordinate, with no notion that a collar shifted upward is the same discriminative cue. Small translations, pose variation, and contrast changes alter many coordinates while the garment category is unchanged.

HOG Descriptors (324-d)

HOG improves on pixels by summarizing local edge orientations. However, it cannot learn that a button-row edge pattern near the neckline is more diagnostic than a similar edge elsewhere. Cell pooling removes exactly the fine spatial cues needed for upper-body distinction.

Linear Decision Boundaries (LogReg, SVM)

A hyperplane cannot create discriminative evidence absent in the input representation. If Shirt and T-shirt/top share overlapping pixel or HOG distributions, changing the margin penalty C cannot resolve the overlap -- this explains the consistent recall plateau across all hyperparameter searches.

5.3 Visual Evidence

- svm_pixels_confusion_matrix.png -- Full 10-class normalized confusion matrix.
- svm_pixels_upper_body_confusion.png -- Focused 4x4 sub-matrix: T-shirt/Pullover/Coat/Shirt.
- svm_pixels_2_as_4_gallery.png -- Gallery: Pullover images misclassified as Coat.
- svm_pixels_6_as_0_gallery.png -- Gallery: Shirt images misclassified as T-shirt/top.

6. Efficiency Analysis

6.1 Accuracy vs. Inference Latency

Model	Features	Latency (ms/img)	Test Accuracy
Logistic Regression	HOG	0.0004	84.55%
Linear SVM	HOG	0.0006	84.23%
Logistic Regression	Pixels	0.0007	84.32%
Linear SVM	Pixels	0.0013	83.80%
Random Forest	Pixels	0.0264	87.43%
Random Forest	HOG	0.0291	84.35%
k-NN	HOG	0.0980	83.84%
k-NN	Pixels	0.2741	85.41%

Table 4. All configurations sorted by inference latency. Linear models are 100-700x faster than k-NN.

6.2 Deployment Guidance

Maximum accuracy: Random Forest on pixels -- 87.43% at 0.026 ms/image.

Real-time throughput (<1 ms budget): Logistic Regression on HOG -- 84.55% at 0.0004 ms/image, with only 3,250 parameters.

Edge deployment: Logistic Regression or SVM on HOG -- 3,250 parameters (vs. 7,850 for pixel variants).

k-NN caveat: Despite 85.41% accuracy on pixels, k-NN stores 179 MiB and is 685x slower than LogReg. Not suitable for production scale.

7. Bridge to Phase 2

The consistent Shirt confusion (mean recall 57.7%) across all 8 Phase 1 configurations is a representation problem, not a hyperparameter problem. A CNN directly addresses each identified limitation:

CNN Property	Phase 1 Limitation Addressed
Convolutions share detectors across locations	Collar/button-row edges are position-sensitive in pixel/HOG space
Stacked layers compose features hierarchically	Fixed HOG cannot integrate edges into semantic garment parts
Pooling provides modest translation tolerance	Pixel flattening is corrupted by small garment position shifts
End-to-end learned features	HOG vocabulary is fixed before seeing the task; cannot specialize

7.1 Phase 2 Success Criteria

- Shirt recall above **62.6%** (the Phase 1 ceiling across all 8 models).
- Reduction in Shirt -> T-shirt/top and Shirt -> Pullover confusion entries in the focused 4x4 upper-body sub-matrix.
- Overall accuracy improvement over the best AML model (87.43%).
- Same held-out test split and metric suite for direct comparability.

8. Conclusion

Dimension	Winner	Value
Best Accuracy	Random Forest + Pixels	87.43%
Best Macro F1	Random Forest + Pixels	87.31%
Fastest Inference	Logistic Regression + HOG	0.0004 ms/img
Most Compact	LogReg / SVM + HOG	3,250 parameters
Hardest Category	**Shirt (class 6)**	**57.7% mean recall**

Table 5. Phase 1 summary results.

Phase 1 establishes a rigorous, fully reproducible 8-configuration AML baseline on Fashion-MNIST. All four models plateau at the same fundamental limit: fine-grained upper-body garment categories require spatial feature composition that fixed descriptors and linear decision boundaries cannot provide. Phase 2 (CNN) will directly test whether learned hierarchical spatial features close the gap that eight Phase 1 configurations could not.

All metrics evaluated on the official Fashion-MNIST held-out test set (10,000 images). Source data: reports/results/.json. Full visualizations: notebooks/00_master_aml_comparison.ipynb.*